

Level 0 SIT: A Physical-Frame Model and Condensate-Source Falsification Contract

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Abstract

This is a model and methods note; it reports no measured anomaly, constraint-cleared parameter region, or validated physical SIT effect. It narrows the Level 0 surface of Super Information Theory to a specified scalar-tensor effective-model contract. The Einstein-frame fields are a metric and two neutral real scalar amplitudes, while ordinary matter follows the single physical metric $\tilde{g}_{mn} = \Omega(\tau, \sigma)^2 g_{mn}$ and electromagnetism remains an independent Maxwell sector. The baseline action by itself does not distinguish a Bose-Einstein condensate from a matched thermal cloud. We therefore add one explicit new hypothesis: the Penrose-Onsager condensate density sources σ through a dimension-four matter operator. In the weak-field point-source regime this gives the signed differential acceleration $\eta_Q = 2\beta_S \dot{g}_Q (f_{\text{BEC}} - f_{\text{thermal}})(1+x)\exp(-x)$, with $x = m_\sigma r$, and the idealized interferometer phase $\Delta \phi_Q = k_{\text{eff}} g \eta_Q T^2$. For a linear, unscreened, uniform spherical source, the point response is replaced by an exact finite-source form factor. The committed regression fixture also propagates first-order parameter uncertainty and five explicitly placeholder apparatus-systematic terms. These calculations test the formula, geometry, and null limits; they are not measured performance, a sensitivity forecast, a constraint-cleared parameter region, or a prediction derived from SIT79. A full source stress tensor, microscopic operator derivation, nonlinear screening calculation, experimental covariance model, and direct likelihood translation remain open. The contribution is a source-explicit linear-response benchmark and falsification contract, not evidence for full SIT or an observed gravity anomaly.

Scope and evidence boundary

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Reader Orientation: The Problem Before the Terminology

Imagine preparing two clouds of the same atomic species in the same apparatus. They have matched atom number, position, launch, geometry, and ordinary composition, but one contains a large Bose-Einstein-condensed component and the other is predominantly thermal. A differential-gravity proposal must say exactly what physical variable distinguishes those preparations, how that variable enters a field equation, and what measurement would change. The word *coherence* cannot do that work by itself.

Established physics supplies several pieces without supplying the proposed effect. Scalar-tensor models specify which metric matter follows and how an extra scalar couples to matter. Screening models show why a light scalar may need nonlinear environmental analysis. Penrose and Onsager define condensation through the largest eigenvalue and mode of the one-body density matrix. Atom interferometers translate acceleration

differences into phase differences. None of those inherited pieces says that condensate density carries a new gravitational charge.

The operation proposed here can therefore be stated in plain language before it is named:

1. reconstruct or measure the condensate mode and condensate fraction;
2. convert that mode into a spatial condensate-density field;
3. declare, as a new hypothesis, that this density sources a neutral scalar;
4. solve the bounded linear field problem for a specified source geometry;
5. compare the accelerations or interferometer phases of matched condensed and thermal preparations; and
6. require the effect to vanish when the new coupling, source charge, or condensate-fraction difference vanishes.

This sequence is the **condensate-source falsification contract**. The broader historical phrase *coherence-source* is retained as lineage, but the measurable input in this paper is the Penrose-Onsager condensate density, not an undefined amount of coherence. If the source operator cannot be completed consistently, if screening removes every accessible effect, if direct constraints eliminate the parameter space, or if calibrated measurements agree with the null, the corresponding model is narrowed or excluded. The contract remains useful because it makes those failure conditions executable.

1. Scope

The v1 paper correctly excluded full SIT, the Coherence Tower, HSIT, RH, neuroscience, holography, Yang-Mills, and particle-physics derivations. This v2 retains those exclusions and closes three ambiguities that prevented a physical calculation: it chooses one matter frame, separates physical electromagnetism from the coherence order parameter, and states the additional matter operator required for a BEC / thermal distinction.

The only accepted thesis is:

A two-scalar extension of Einstein-Maxwell-matter theory plus a declared condensate-density source operator yields a bounded weak-field differential-acceleration formula. Whether the operator has a microscopic origin or any nonzero viable parameter region exists remains open.

2. Fields and Physical Metric

Use four-dimensional natural units. The model contains:

- an Einstein-frame metric g_{mn} ;
- a real scalar τ ;
- a real macroscopic amplitude σ ;
- an independent electromagnetic connection A_m with curvature F_{mn} ; and
- ordinary matter fields χ_i .

Matter rods and clocks couple to

$$g_{\tilde{mn}} = \Omega(\tau, \sigma)^2 g_{mn}. \quad (1)$$

This Jordan metric is the physical metric by definition. Observable time, length, acceleration, stress-energy, and apparatus geometry must be evaluated in this frame or translated to it consistently.

The wider SIT program may retain a complex order parameter and phase language. The Level 0 benchmark does not identify that phase connection with physical electromagnetism. If the coherence condensate were minimally charged under the electromagnetic $U(1)$, a nonzero vacuum amplitude would generically raise a

photon-mass question. This letter avoids that unexamined claim by setting the physical electric charge of σ to zero. A future hidden $U(1)$ would be a separate model with separate constraints.

3. Benchmark Action

The action is

$$\begin{aligned}
S = \int d^4x \sqrt{-g} [& \\
& M_{Pl}^2 R/2 \\
& - (\partial_\tau \sigma)^2/2 \\
& - (\partial_\sigma \sigma)^2/2 \\
& - V(\tau, \sigma) \\
& - f_{EM}(\tau, \sigma) F_{mn} F^{mn}/4 \\
&] \\
& + S_m[\Omega(\tau, \sigma)^2 g, A, \chi].
\end{aligned} \tag{2}$$

Einstein-Hilbert gravity and Maxwell theory are assumed sectors. Their recovery in a limit is not a derivation of their origin.

This baseline action has no coherence-specific matter source. The Draft 5 laboratory extension therefore adds

$$L_Q = -(g_Q m_a/M_{Pl}) \sigma n_0. \tag{3}$$

where m_a is the atomic mass and n_0 is the condensate number density in the physical apparatus frame. This is a new modeling hypothesis, not a consequence of SIT79. Since $[\sigma]=1$ and $[n_0]=3$, the interaction has mass dimension four. A covariant completion must represent $n_0 = -u_m j_0^m$ with the apparatus four-velocity, conserve the condensate current during the modeled interval, and include its stress tensor and exchange terms.

A first stable polynomial specification is

$$\begin{aligned}
V = & m_\tau^2 \tau^2/2 \\
& + m_\sigma^2 \sigma^2/2 \\
& + m_{link}^2 (\tau - \alpha \sigma)^2/2 \\
& + \lambda_\tau \tau^4/4 \\
& + \lambda_\sigma \sigma^4/4,
\end{aligned}$$

$$\Omega = \exp(\beta_\tau \tau/M_{Pl} + \beta_\sigma \sigma/M_{Pl}),$$

$$f_{EM} = 1 + \xi_\tau \tau/M_{Pl} + \xi_\sigma \sigma/M_{Pl} + \text{higher terms}. \tag{4}$$

Every mass, coupling, cutoff, higher operator, and boundary condition is model data. Positive quadratic masses and quartics supply a stable local origin for the committed recovery benchmark; they do not establish radiative stability or global UV consistency.

4. Variation Convention

Define the physical-frame matter tensor by

$$\begin{aligned}
\delta S_m = & (1/2) \int d^4x \sqrt{-g} \\
& T^{\tilde{mn}} \delta g_{\tilde{mn}}
\end{aligned} \tag{5}$$

Since $\delta \tilde{g}_{mn} = 2 \tilde{g}_{mn} \delta \ln \Omega$, the matter variation with respect to either scalar is proportional to the physical trace T and the corresponding derivative of $\ln \Omega$. With the displayed action, the scalar equations have the structural form

$$\begin{aligned} \Box \tau - V_{,\tau} - f_{EM,\tau} F^2/4 \\ + \Omega^4 T (\ln \Omega)_{,\tau} = 0, \end{aligned}$$

$$\begin{aligned} \Box \sigma - V_{,\sigma} - f_{EM,\sigma} F^2/4 \\ + \Omega^4 T (\ln \Omega)_{,\sigma} = 0. \end{aligned} \quad (6)$$

under the stated stress-tensor convention. The internal full-variation audit gives

$$\begin{aligned} \Box \sigma - V_{,\sigma} - f_{EM,\sigma} F^2/4 \\ + \Omega^4 T (\ln \Omega)_{,\sigma} + J_Q = 0, \end{aligned} \quad (7)$$

$$J_Q = (1/\sqrt{-g}) \delta S_Q / \delta \sigma.$$

For the shorthand source action

$$S_Q = - \int \sqrt{-g} (g_Q m_a / M_{Pl}) \sigma \, d^4x. \quad (8)$$

holding n_0 fixed during scalar variation gives $J_Q = -(g_Q m_a / M_{Pl}) n_0$. The corresponding Maxwell and structural metric equations are

$$\nabla_\mu (f_{EM} F^{\mu\nu}) = J_{matter}^\nu,$$

$$\begin{aligned} M_{Pl}^2 G_{mn} = T_{\tau mn} + T_{\sigma mn} + f_{EM} T_{EM mn} \\ + T_{matter mn} + T_{Q mn}. \end{aligned} \quad (9)$$

Here $T_{Q mn} = -2 \delta S_Q / (\sqrt{-g} \delta g^{mn})$ is not yet determined. Specifying it requires the independent condensate variables, current and four-velocity normalization, pressure and gradient energy, trapping terms, and exchange with the noncondensed fraction. The Bianchi identity applies to the combined stress tensor, so the shorthand source cannot simultaneously be treated as fixed and promoted to a complete covariant matter theory. Draft 5 therefore uses it only as an external fixed-source approximation with omitted backreaction. The scalar signs, Maxwell equation, and structural metric equation have been derived internally under the declared convention; complete source stress / current conservation and independent review remain open.

For the bounded source benchmark, decouple τ , set $f_{EM}=1$, use $\Omega = \exp(\beta_\sigma \sigma / M_{Pl})$, and linearize around $\sigma=0$. With nonrelativistic density ρ and the new source term, the static equation is

$$\begin{aligned} (\nabla^2 - m_\sigma^2) \sigma \\ = \beta_\sigma \rho / M_{Pl} + g_Q m_a n_0 / M_{Pl}. \end{aligned} \quad (10)$$

This linear equation fixes the sign convention used below. It does not replace a covariant source action, nonlinear screening solution, or source-sector conservation review.

5. Null Recovery Benchmark

The machine-readable null benchmark sets

$$\begin{aligned} \beta_\tau &= \beta_\sigma = \xi_\tau = \xi_\sigma = 0, \\ \Omega &= 1, \\ f_{EM} &= 1. \end{aligned} \quad (11)$$

Its quadratic scalar mass matrix is

$$\begin{bmatrix} [m_{\text{tau}}^2 + m_{\text{link}}^2, & -\alpha m_{\text{link}}^2], \\ [-\alpha m_{\text{link}}^2, m_{\text{sigma}}^2 + \alpha^2 m_{\text{link}}^2] \end{bmatrix}. \quad (12)$$

The committed parameters give two positive eigenvalues, as checked by `model/validate_benchmark.py`. The benchmark therefore supplies a stable local scalar vacuum and decoupled Einstein-Maxwell-matter recovery at the classical quadratic level. It makes no prediction for $\Delta g/g$ and is not evidence of a coherence-gravity interaction.

6. A Declared Condensate-to-Source Map

The symbol σ remains a classical model amplitude. It is not condensate fraction, interference visibility, off-diagonal long-range order, density-matrix purity, or phase-correlation length. Draft 5 instead defines a separate condensate observable from the one-body reduced density matrix:

$$\begin{aligned} \text{integral } \rho_1(x, x') \phi_0(x') d^3x' &= N_0 \phi_0(x), \\ \text{integral } |\phi_0(x)|^2 d^3x &= 1, \\ n_0(x) &= N_0 |\phi_0(x)|^2, \\ f_0 &= N_0/N. \end{aligned} \quad (13)$$

N_0 is the largest Penrose-Onsager eigenvalue and ϕ_0 its normalized mode. The map uses the measured or reconstructed condensate fraction; it does not force the thermal preparation to have exactly zero f_0 .

The resulting operational chain is

$$\begin{aligned} Q: \{ \rho_1, \text{apparatus resolution} \} \\ \rightarrow \{ N_0, \phi_0, n_0, f_0 \} \\ \rightarrow \{ \text{source term } (g_Q m_a/M_{\text{Pl}}) n_0 \}. \end{aligned} \quad (14)$$

This closes the benchmark definition, not its microscopic justification. The new coupling must still account for preparation energy, ordinary stress-energy, finite-temperature fluctuations, source backreaction, and the experimental method used to estimate ρ_1 and f_0 .

7. Nonzero Linear-Response Result and Constraint Program

Use the reduced Planck convention $M_{\text{Pl}}^{-2} = 8\pi G$. For a point source with effective scalar coupling β_S , define

$$Y(x) = (1+x) \exp(-x), \quad x = m_{\text{sigma}} r. \quad (15)$$

A cloud with condensate fraction f_0 has effective scalar coupling

$$\beta_{\text{cloud}} = \beta_{\text{sigma}} + g_Q f_0. \quad (16)$$

The scalar acceleration relative to Newtonian gravity is therefore

$$a_{\text{sigma}}/a_N = 2 \beta_S (\beta_{\text{sigma}} + g_Q f_0) Y(x). \quad (17)$$

For matched preparations, the universal term cancels and the declared coherence-source operator gives

$$\begin{aligned} \eta_Q &= (a_{\text{BEC}} - a_{\text{thermal}})/a_N \\ &= 2 \beta_S g_Q (f_{\text{BEC}} - f_{\text{thermal}}) Y(x). \end{aligned} \quad (18)$$

This is a derived result of the stated linear-response benchmark. It is not a derivation of g_Q , and it is not constraint-cleared. Its sign follows $\beta_S g_Q$; it vanishes for $g_Q=0$, $\beta_S=0$, equal condensate fractions, or short scalar range.

Existing MICROSCOPE, dual-isotope, and coherent-internal-state tests are neighboring constraints, not direct BEC-versus-thermal bounds on this operator. A constraint fit must derive the translation before using their reported sensitivities. It must also include finite Earth / source form factors, screening, inverse-square-law and PPN limits, electromagnetic effects if x_i is nonzero, and the source-sector stress-energy.

If no allowed region remains, the source-coupled benchmark is excluded. If no direct translation can be justified, the paper remains a model and methods note rather than a prediction letter.

7.1 Exact Linear Uniform-Sphere Benchmark

For a uniform spherical source of radius R , total mass M , inverse scalar range μ , and exterior detector radius $r=R+h$, define $z=\mu R$ and $y=\mu h$. The exact linear unscreened source-amplitude factor is

$$A(z)=3[z \cosh(z)-\sinh(z)]/z^3,$$

$$Y_{\text{sphere}}(z,y)=A(z)[1+z+y]\exp(-z-y). \quad (19)$$

The implementation evaluates the algebraically equivalent stable form

$$Y_{\text{sphere}}(z,y)=3(1+z+y)/(2z^3) [(z-1)\exp(-y)+(z+1)\exp(-2z-y)]. \quad (20)$$

and uses the small- z series $A(z)=1+z^2/10+z^4/280+\dots$. For the declared fixture $R=6,371,000$ m, $h=100$ m, and $\mu(R+h)=0.1$, the point response is 0.9953211598395556, the uniform-sphere response is 0.9963168052700276, and their ratio is 1.001000325794975. With the illustrative coupling fixture this gives $\eta_Q=1.5542542162212432e-12$ and $\Delta\phi_Q=2.456440917215015e-4$ rad.

This is exact only for a linear, unscreened, uniform sphere. It is not a layered Earth model, laboratory source geometry, thin-shell solution, or nonlinear screening calculation. The numbers are deterministic software regressions, not measurements, forecasts, sensitivities, allowed regions, or physical validation.

8. BEC Versus Thermal Protocol

The experiment remains a falsifier architecture, not a constraint-cleared prediction. It compares the same atomic species under matched atom number, launch, trap, geometry, external fields, and readout, with deliberately different measured condensate fractions.

For an idealized light-pulse interferometer,

$$\Delta\phi_Q = k_{\text{eff}} g \eta_Q T^2. \quad (21)$$

The committed regression point uses explicit β_S , g_Q , condensate fractions, source geometry and range, laser wavelength, and interrogation time. Centered numerical differentiation propagates declared standard uncertainties in wavelength, interrogation time, local gravity, both condensate fractions, source radius, and detector altitude. It gives a parameter-only standard uncertainty of $7.042182760266661e-6$ rad.

Five separately declared phase-systematic placeholders cover wavefront aberration, Coriolis / rotation, gravity-gradient / launch mismatch, AC-Stark and Zeeman effects, and detection / fit bias. Their quadrature is $2.2360679774997898e-4$ rad; combining it with the parameter term gives $2.2371766210567507e-4$ rad. The resulting illustrative signal-to-placeholder ratio is 1.098009381152344. Because these systematic

values are placeholders rather than calibrated distributions, none of these numbers is an experimental forecast, sensitivity, significance, exclusion, or allowed parameter claim.

Required ordinary controls include density profile, interaction energy, temperature, condensate fraction, release dynamics, magnetic and electric fields, gravity gradients, vibrations, imaging bias, and decoherence. A blinded analysis must evaluate a preregistered differential acceleration or phase statistic.

The complete forward model must replace the uniform source by the real Earth or laboratory density profile, establish or solve the screening regime, propagate both clouds through the pulse sequence, and fit measured controls and covariances jointly. A null result excludes only parameter points whose validated signal exceeds demonstrated sensitivity under a justified likelihood. A positive residual supports only the specified source-coupled model relative to the nulls and alternatives; it does not prove full SIT.

9. Corrected External Anchors

The external literature supplies constraints and experimental capability, not validation of this model.

- Khoury and Weltman and Hinterbichler and Khoury provide examples of screened scalar mechanisms whose assumptions must be derived for a benchmark.
- MICROSCOPE reports a stringent null test of the weak equivalence principle.
- Penrose and Onsager define condensation through an extensive largest eigenvalue of the one-body reduced density matrix; Draft 5 uses that definition for N_0 and f_0 .
- Asenbaum et al. constrain dual-isotope free fall, and Rosi et al. test atoms in internal-state superpositions. Neither paper directly constrains the new condensate-fraction operator without a translation calculation.
- Cold-atom and BEC interferometry demonstrates relevant preparation and metrology capabilities.
- The space-interferometry citation is correctly identified as Maike D. Lachmann, Holger Ahlers, Dennis Becker, et al., *Nature Communications* 12, article 1317 (2021), DOI 10.1038/s41467-021-21628-z; v1's author / article metadata was incorrect.

10. Dependency and Claim Boundary

Paper 13 may use the declared map, source operator, sign convention, linear-response formula, and exact uniform-sphere fixture as private theory inputs. It must remain a feasibility and protocol paper until Level 0 supplies a realistic source profile, screening calculation, constraint-cleared parameter region, and calibrated apparatus / covariance model. Paper 16 may record source and hash provenance but cannot validate the physics. Branch C finite formal artifacts do not derive this action or its observables.

The Level 0 paper makes no claim about physical SIT, full SIT, HSIT, the Coherence Tower, SIT / Riemann, RH, Yang-Mills, P versus NP, SAN, or consciousness.

11. Release and Promotion Gates

Version 3 fixes the endpoint as a model-and-methods note. The exact linear uniform-sphere solution, first-order uncertainty propagation, and placeholder systematics are retained as reproducible software and bookkeeping tests. They are not promoted to a sensitivity forecast, viable parameter point, or experimental prediction.

For this bounded release, the citation, equation, cross-reference, source-to-PDF, rendering, privacy, package, metadata, and version-lineage checks are local acceptance gates. Independent scalar-tensor, many-body,

cold-atom, apparatus, and statistical review remains recommended and may motivate a later version; it is not classified as unfinished local drafting work for this exact model-and-methods preprint.

Any later prediction or experimental-outreach version additionally requires a covariant source completion or justified fixed-source domain, realistic source profile, nonlinear screening analysis, direct constraint likelihood, calibrated apparatus transfer functions and covariances, and a blinded analysis plan. Those are promotion gates, not results of this note.

12. Conclusion

The Level 0 model-and-methods note contains a precise declared reason why a BEC and thermal cloud could differ: a new condensate-density source operator defined through the Penrose-Onsager condensate mode. The null benchmark verifies the recovery path; the nonzero linear response verifies the signed Yukawa and interferometer formulas; and the exact linear uniform-sphere and first-order uncertainty fixtures test source geometry and apparatus bookkeeping without manufacturing a constraint-cleared anomaly. The internal variation audit also exposes the remaining covariant source stress / current problem instead of hiding it. The decisive question is whether the operator has a defensible microscopic origin and survives realistic source, screening, fifth-force, cold-atom, and calibrated apparatus constraints. Until those gates and independent reviews close, the paper is a model and falsification contract rather than a validated prediction letter. Its publication-level result is the explicit source hypothesis, linear-response derivation, exact bounded fixture, reproducibility surface, and failure criteria - not evidence that the proposed interaction exists.

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Approval Gate

This Version 3 final preprint is locally complete. No Zenodo version, OSF change, GitHub push, journal submission, or experimental outreach action has yet been performed. Publication follows the governed portfolio release sequence. Independent specialist review remains recommended and may motivate a later version; it is not a condition for classifying this exact manuscript as the completed Version 3 preprint.